Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Infinite Nested Radical Integral

Evaluate the definite integral:

$$\int_0^1 \sqrt{x + \sqrt{x^2 + \sqrt{x^4 + \sqrt{x^8 + \cdots}}}}\, dx$$

Solution: Let $S(x)$ denote the infinite nested radical function inside the integral:

$$S(x) = \sqrt{x + \sqrt{x^2 + \sqrt{x^4 + \sqrt{x^8 + \cdots}}}}$$

We can exploit the self-similar structure of this nested radical by factoring out the powers of x. Factoring x^2 from the second radical, x^4 from the third, and so on, allows us to write:

$$S(x) = \sqrt{x + x\sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}$$

The inner infinite sequence of nested radicals is composed purely of 1s. Let $C = \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}$. Squaring both sides yields $C^2 = 1 + C$, which is the characteristic equation for the Golden Ratio. Solving $C^2 - C - 1 = 0$ for the positive root gives $C = \phi = \frac{1+\sqrt{5}}{2}$. Substituting this back into our expression for $S(x)$:

$$S(x) = \sqrt{x + x\phi} = \sqrt{x(1+\phi)}$$

Using the fundamental property of the Golden Ratio, $1 + \phi = \phi^2$, the function greatly simplifies to:

$$S(x) = \sqrt{x\phi^2} = \phi\sqrt{x}$$

Now we can seamlessly evaluate the integral:

$$\int_0^1 \phi\sqrt{x}\, dx = \phi\left[\frac{2}{3}x^{3/2}\right]_0^1 = \frac{2}{3}\phi$$

Substituting the exact value of ϕ:

$$\frac{2}{3}\left(\frac{1+\sqrt{5}}{2}\right) = \frac{1+\sqrt{5}}{3}$$

1.2 Hard Integral

Improper Inverse Trigonometric Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \left[\frac{\pi}{2} - \tan^{-1}\left(x^2 + \frac{1}{x^2}\right)\right] dx$$

Solution: Let the integral be I. Using the identity $\frac{\pi}{2} - \tan^{-1}(u) = \tan^{-1}\left(\frac{1}{u}\right)$ for $u > 0$, we rewrite the integrand:

$$I = \int_{-\infty}^{\infty} \tan^{-1}\left(\frac{1}{x^2 + x^{-2}}\right) dx = 2\int_{0}^{\infty} \tan^{-1}\left(\frac{1}{x^2 + x^{-2}}\right) dx$$

Let $I_0 = \frac{I}{2}$. We apply the substitution $x \to \frac{1}{x}$, which gives $dx \to -\frac{1}{x^2}dx$:

$$I_0 = \int_{0}^{\infty} \frac{1}{x^2} \tan^{-1}\left(\frac{1}{x^2 + x^{-2}}\right) dx$$

Adding our original expression for I_0 to this newly substituted version gives:

$$2I_0 = \int_{0}^{\infty} \left(1 + \frac{1}{x^2}\right) \tan^{-1}\left(\frac{1}{x^2 + x^{-2}}\right) dx$$

Next, we introduce the substitution $u = x - \frac{1}{x}$, which means $du = \left(1 + \frac{1}{x^2}\right) dx$. Note that $u^2 + 2 = x^2 + x^{-2}$. As x ranges from 0 to ∞, u maps from $-\infty$ to ∞. Thus, the integral transforms into:

$$2I_0 = \int_{-\infty}^{\infty} \tan^{-1}\left(\frac{1}{u^2 + 2}\right) du \implies I_0 = \int_{0}^{\infty} \tan^{-1}\left(\frac{1}{u^2 + 2}\right) du$$

Apply integration by parts on I_0 using $w = \tan^{-1}\left(\frac{1}{u^2+2}\right)$ and $dv = du$:

$$I_0 = \left[u \tan^{-1}\left(\frac{1}{u^2 + 2}\right)\right]_0^{\infty} - \int_{0}^{\infty} u \left(\frac{1}{1 + \frac{1}{(u^2+2)^2}}\right) \left(\frac{-2u}{(u^2 + 2)^2}\right) du$$

The boundary term evaluates to zero. Simplifying the remaining integrand:

$$I_0 = \int_{0}^{\infty} \frac{2u^2}{(u^2 + 2)^2 + 1} du = \int_{0}^{\infty} \frac{2u^2}{u^4 + 4u^2 + 5} du$$

This takes the form of the well-known integral identity $\int_0^{\infty} \frac{x^2}{x^4+bx^2+c} dx = \frac{\pi}{2\sqrt{b+2\sqrt{c}}}$. Here, $b = 4$ and $c = 5$:

$$I_0 = 2\left(\frac{\pi}{2\sqrt{4 + 2\sqrt{5}}}\right) = \frac{\pi}{\sqrt{4 + 2\sqrt{5}}}$$

Because $I = 2I_0$:

$$I = \frac{2\pi}{\sqrt{4 + 2\sqrt{5}}}$$

By noting that $(4 + 2\sqrt{5})(2\sqrt{5} - 4) = 4$, we find that $\frac{1}{\sqrt{4+2\sqrt{5}}} = \frac{\sqrt{2\sqrt{5}-4}}{2}$. Thus, the final form simplifies nicely:

$$I = \pi\sqrt{2\sqrt{5} - 4}$$

Differentials

2.1 Medium Differential

Second-Order Linear Initial Value Problem

Solve the second-order linear differential equation:

$$y'' - 10y' + 25y = 0$$

Given the initial conditions $y(0) = 4$ and $y'(0) = 17$, find the exact numerical value of the expression $y(2) \cdot e^{-10}$.

Solution: We begin by setting up the characteristic equation for the differential equation:

$$r^2 - 10r + 25 = 0 \implies (r-5)^2 = 0$$

Since we have a repeated root $r = 5$, the general solution for $y(x)$ takes the form:

$$y(x) = (A + Bx)e^{5x}$$

We apply the first initial condition, $y(0) = 4$:

$$y(0) = (A + B(0))e^0 = A = 4$$

Next, we take the derivative of the general solution to apply the second initial condition:

$$y'(x) = Be^{5x} + 5(A + Bx)e^{5x}$$

Evaluate this at $x = 0$ and equate it to 17:

$$y'(0) = B + 5A = 17$$

Substitute $A = 4$ into the equation:

$$B + 5(4) = 17 \implies B = 17 - 20 = -3$$

The particular solution is therefore:

$$y(x) = (4 - 3x)e^{5x}$$

We want to evaluate the expression $y(2) \cdot e^{-10}$. First, find $y(2)$:

$$y(2) = (4 - 3(2))e^{5(2)} = (4 - 6)e^{10} = -2e^{10}$$

Finally, multiply by e^{-10}:

$$y(2) \cdot e^{-10} = (-2e^{10}) \cdot e^{-10} = -2$$

2.2 Hard Differential

Laplace Transform Differential Equation

The Change Hog engine parses continuous commit histories across x parallel nodes. Telemetry indicates that the relative rate of change of the parsing latency L (in ms) with respect to the node count is continuously equal to the product of $10x$ and the improper integral from 0 to ∞ of $e^{-xt}\sin(t)$ with respect to t. If the baseline latency of the system is exactly 2 ms at $x = 0$ nodes, find the exact latency in milliseconds at $x = 1$ node.

Solution: The differential equation models the rate of change of the parsing latency $L(x)$:

$$\frac{dL}{dx} = 10x \int_0^\infty e^{-xt} \sin(t)\, dt$$

The improper integral is structurally identical to the Laplace transform of $\sin(t)$, evaluated at the parameter $s = x$. Recall the standard Laplace transform formula:

$$\mathcal{L}\{\sin(t)\}(x) = \int_0^\infty e^{-xt} \sin(t)\, dt = \frac{1}{x^2+1}$$

Substituting this back, our differential equation drastically simplifies:

$$\frac{dL}{dx} = \frac{10x}{x^2+1}$$

We now integrate both sides with respect to x to find $L(x)$:

$$L(x) = \int \frac{10x}{x^2+1}\, dx = 5\ln(x^2+1) + C$$

The problem provides a baseline latency of 2 ms at $x = 0$ nodes, which serves as our initial condition $L(0) = 2$:

$$L(0) = 5\ln(0^2+1) + C = 5\ln(1) + C = 2 \implies C = 2$$

Thus, the exact latency function is:

$$L(x) = 5\ln(x^2+1) + 2$$

We evaluate this function at $x = 1$ node to find the requested latency:

$$L(1) = 5\ln(1^2+1) + 2 = 5\ln(2) + 2$$

Derivatives

3.1 Medium Derivative

Infinite Sine Series Exponential

Let $f(x) = e^{\left(\sin(x)-\sin\left(\frac{x}{3}\right)+\sin\left(\frac{x}{5}\right)-\sin\left(\frac{x}{7}\right)+\dots\right)}$. Find $f'(0)$.

Solution: Let the exponent be represented by a function $g(x)$:

$$g(x) = \sin(x) - \sin\left(\frac{x}{3}\right) + \sin\left(\frac{x}{5}\right) - \sin\left(\frac{x}{7}\right) + \cdots = \sum_{k=0}^{\infty}(-1)^k \sin\left(\frac{x}{2k+1}\right)$$

By the chain rule, the derivative of $f(x) = e^{g(x)}$ is:

$$f'(x) = g'(x)e^{g(x)}$$

To find $f'(0)$, we must determine $g(0)$ and $g'(0)$. Clearly, evaluating the sine series at $x = 0$ forces every term to zero, yielding $g(0) = 0$. Now, take the term-by-term derivative of $g(x)$:

$$g'(x) = \cos(x) - \frac{1}{3}\cos\left(\frac{x}{3}\right) + \frac{1}{5}\cos\left(\frac{x}{5}\right) - \frac{1}{7}\cos\left(\frac{x}{7}\right) + \dots$$

Evaluate $g'(x)$ at $x = 0$. Since $\cos(0) = 1$ for all terms:

$$g'(0) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

This alternating sum is the well-known Gregory-Leibniz series, which converges to $\frac{\pi}{4}$. Thus, $g'(0) = \frac{\pi}{4}$. Finally, piece everything together to compute $f'(0)$:

$$f'(0) = g'(0)e^{g(0)} = \frac{\pi}{4}e^0 = \frac{\pi}{4}$$

3.2 Hard Derivative

Infinite Rational Series Derivative

Given the function $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^n}{x^2+n}$. Find $f''(1)$.

Solution: First, take the first derivative of the given sum term-by-term with respect to x:

$$f'(x) = \sum_{n=1}^{\infty}(-1)^n \frac{-2x}{(x^2+n)^2}$$

Apply the quotient rule to take the second derivative:

$$f''(x) = \sum_{n=1}^{\infty}(-1)^n \left[\frac{-2(x^2+n)^2 - (-2x)(2)(x^2+n)(2x)}{(x^2+n)^4}\right]$$

Cancel a factor of (x^2+n) from the numerator and denominator:

$$f''(x) = \sum_{n=1}^{\infty}(-1)^n \frac{-2(x^2+n)+8x^2}{(x^2+n)^3} = \sum_{n=1}^{\infty}(-1)^n \frac{6x^2-2n}{(x^2+n)^3}$$

Evaluate this series specifically at $x=1$:

$$f''(1) = \sum_{n=1}^{\infty}(-1)^n \frac{6-2n}{(n+1)^3}$$

Let us shift the index to $k=n+1$, meaning $n=k-1$. As n traverses from 1 to ∞, k runs from 2 to ∞:

$$f''(1) = \sum_{k=2}^{\infty}(-1)^{k-1}\frac{6-2(k-1)}{k^3} = \sum_{k=2}^{\infty}(-1)^{k-1}\frac{8-2k}{k^3} = 8\sum_{k=2}^{\infty}\frac{(-1)^{k-1}}{k^3} - 2\sum_{k=2}^{\infty}\frac{(-1)^{k-1}}{k^2}$$

These sums relate directly to the Dirichlet eta function $\eta(s) = \sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k^s} = (1-2^{1-s})\zeta(s)$. For $s=3$:

$$\eta(3) = \frac{3}{4}\zeta(3) \implies \sum_{k=2}^{\infty}\frac{(-1)^{k-1}}{k^3} = \frac{3}{4}\zeta(3)-1$$

For $s=2$:

$$\eta(2) = \frac{1}{2}\zeta(2) = \frac{\pi^2}{12} \implies \sum_{k=2}^{\infty}\frac{(-1)^{k-1}}{k^2} = \frac{\pi^2}{12}-1$$

Substituting these known values back into our formula for $f''(1)$:

$$f''(1) = 8\left(\frac{3}{4}\zeta(3)-1\right) - 2\left(\frac{\pi^2}{12}-1\right) = 6\zeta(3) - 8 - \frac{\pi^2}{6} + 2$$

$$f''(1) = 6\zeta(3) - \frac{\pi^2}{6} - 6$$

Physics & Engineering

4.1 Mechanics

Elastic Collision of a Rolling Sphere

A solid sphere rolling on a rough horizontal surface with a linear speed $v = 14\ \mathrm{m\,s^{-1}}$ collides elastically with a fixed smooth, vertical wall. Find the speed of the sphere in $\mathrm{m\,s^{-1}}$ after it has started pure rolling in the backward direction.

Solution: Assume the right is the positive direction. Before the collision, the sphere rolls purely to the right, so its translational velocity is $v_1 = 14\ \mathrm{m\,s^{-1}}$ and its angular velocity is $\omega_1 = -14/R$ (clockwise). Because the collision is perfectly elastic and the vertical wall is smooth (no vertical impulse), the translational velocity flips to $v_2 = -14\ \mathrm{m\,s^{-1}}$, while the angular velocity ω_2 remains completely unaltered at $-14/R$. The sphere is now slipping. We can determine the final purely rolling state by conserving angular momentum about an axis lying on the ground, thereby making the torque from kinetic friction mathematically vanish. The initial angular momentum about a point on the floor immediately after collision is:

$$L_i = I_{cm}\omega_2 + Mv_2R = \left(\frac{2}{5}MR^2\right)\left(-\frac{14}{R}\right) + M(-14)R = -\frac{28}{5}MR - 14MR = -\frac{98}{5}MR$$

Wait, this is if we consider the appropriate cross products. Taking the standard counter-clockwise definition for the z-axis:

$$\vec{L}_i = \left(\frac{2}{5}MR^2\right)\left(-\frac{14}{R}\hat{k}\right) + (R\hat{\jmath}) \times (-14M\hat{\imath}) = -\frac{28}{5}MR\hat{k} + 14MR\hat{k} = \frac{42}{5}MR\hat{k}$$

When pure rolling is eventually re-established moving to the left, the sphere moves at some final velocity $\vec{v}_f = v\hat{\imath}$ (where $v < 0$) and spins counter-clockwise to match, $\vec{\omega}_f = -v/R\hat{k}$:

$$\vec{L}_f = \left(\frac{2}{5}MR^2\right)\left(-\frac{v}{R}\hat{k}\right) + (R\hat{\jmath}) \times (Mv\hat{\imath}) = -\frac{2}{5}MRv\hat{k} - MRv\hat{k} = -\frac{7}{5}MRv\hat{k}$$

By equating $\vec{L}_i$ and $\vec{L}_f$, we solve for v:

$$\frac{42}{5}MR = -\frac{7}{5}MRv \implies 42 = -7v \implies v = -6\ \mathrm{m\,s^{-1}}$$

The absolute speed of the sphere is exactly $6\ \mathrm{m\,s^{-1}}$.

4.2 Quantum Mechanics

1D Box Ground State Energy

An electron is confined to move inside a rigid one-dimensional box of length $L = 0.2$ nm. Calculate the ground state energy of the electron in electron-volts (eV). Take $h = 6.634 \times 10^{-34}$ J s, mass $m = 9.1 \times 10^{-31}$ kg, and 1 eV $= 1.602 \times 10^{-19}$ J.

Solution: The quantized energy levels for an electron in a 1D rigid box are given by:

$$E_n = \frac{n^2 h^2}{8mL^2}$$

For the ground state, $n = 1$. Let us systematically plug in the given values: $h^2 = (6.634 \times 10^{-34})^2 \approx 44.01 \times 10^{-68}\ \mathrm{J}^2 \cdot \mathrm{s}^2$ $L = 0.2 \times 10^{-9}\ \mathrm{m} \implies L^2 = 0.04 \times 10^{-18} = 4 \times 10^{-20}\ \mathrm{m}^2$ The denominator expands to:

$$8mL^2 = 8(9.1 \times 10^{-31})(4 \times 10^{-20}) = 291.2 \times 10^{-51}\ \mathrm{kg} \cdot \mathrm{m}^2$$

Dividing the numerator by the denominator to get the energy in Joules:

$$E_1 = \frac{44.01 \times 10^{-68}}{291.2 \times 10^{-51}} \approx 0.15113 \times 10^{-17}\ \mathrm{J} = 15.113 \times 10^{-19}\ \mathrm{J}$$

Finally, convert this physical quantity to electron-volts by dividing by the elementary charge equivalence:

$$E_1 = \frac{15.113 \times 10^{-19}\ \mathrm{J}}{1.602 \times 10^{-19}\ \mathrm{J/eV}} \approx 9.43\ \mathrm{eV}$$

4.3 Electric Fields

Submerged Cylindrical Capacitor Current

An air-filled cylindrical capacitor is connected to a DC voltage source of $V = 180$ V. It is then submerged vertically into a tank of water at a constant speed $v = 6.0\ \mathrm{mm\ s}^{-1}$. The gap between the electrodes is $d = 2.0$ mm, and the mean radius is $r = 50$ mm. Assuming $d \ll r$, calculate the current flowing through the lead wires during submersion in μA. Take $\varepsilon = 81$ and $\varepsilon_0 = \frac{1}{36\pi \times 10^9}\ \mathrm{F\ m}^{-1}$.

Solution: Because $d \ll r$, we can approximate the cylindrical setup as a parallel-plate capacitor of active width $2\pi r$. Let $y(t)$ denote the submerged depth of the capacitor over time. The system acts as two parallel capacitors: a water-filled one with height y, and an air-filled one with height $H - y$. The total capacitance $C(t)$ is:

$$C(t) = \frac{\varepsilon\varepsilon_0(2\pi r)y}{d} + \frac{\varepsilon_0(2\pi r)(H - y)}{d} = \frac{2\pi r\varepsilon_0}{d}[y(\varepsilon - 1) + H]$$

The current I drawn from the source is the time derivative of the charge $Q = C(t)V$:

$$I = \frac{dQ}{dt} = V\frac{dC}{dt}$$

Differentiating the capacitance function with respect to t ($\frac{dy}{dt} = v$):

$$\frac{dC}{dt} = \frac{2\pi r\varepsilon_0}{d}(\varepsilon - 1)v$$

Now, we systematically insert the given parameters: $r = 0.05$ m, $d = 0.002$ m, $v = 0.006$ m/s, and $\varepsilon - 1 = 80$:

$$\frac{dC}{dt} = \frac{2\pi(0.05)}{0.002} \times \frac{1}{36\pi \times 10^9} \times 80 \times 0.006$$

$$= (50\pi) \times \frac{1}{36\pi \times 10^9} \times 0.48 = \frac{24\pi}{36\pi \times 10^9} = \frac{2}{3} \times 10^{-9}\ \mathrm{F\,s^{-1}}$$

Multiply this rate of change by the voltage $V = 180$ V to deduce the current:

$$I = 180\left(\frac{2}{3} \times 10^{-9}\right) = 120 \times 10^{-9}\ \mathrm{A} = 0.12\ \mu\mathrm{A}$$